

SYNTHETIC TEST FIXTURE - NOT A REAL PUBLICATION

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A Modified Fuzzy C-Means Algorithm for Bias Field Estimation
and Segmentation of MRI Data
(synthetic stand-in)

Abstract

We present a modification of the fuzzy c-means objective function that compensates for intensity inhomogeneity and allows the labelling of a pixel to be influenced by its immediate neighbourhood. The neighbourhood term acts as a regularizer and biases the solution toward piecewise homogeneous labelling.

2. Method

The standard fuzzy c-means objective is modified by adding a neighbourhood regularization term. The objective function is

$$J = \sum_i \sum_k u_{ik}^p \|y_k - v_i\|^2 + (\alpha / N_R) \sum_i \sum_k u_{ik}^p \sum_{\{y_r \in N_k\}} \|y_r - v_i\|^2$$

where u_{ik} is the membership of voxel k in class i , v_i is the centroid of class i , N_k is the set of neighbours of voxel k , and N_R is the cardinality of that neighbourhood.

We set the neighbourhood weight $\alpha = 0.7$ in all experiments reported here. The fuzzifier is fixed at $p = 2$, which is the conventional choice.

2.1 Optimization

Minimizing J with respect to the memberships gives the update

$$u_{ik} = 1 / \sum_j (D_{ik} / D_{jk})^{1/(p-1)}$$

where $D_{ik} = \|y_k - v_i\|^2 + (\alpha / N_R) \sum_{\{y_r \text{ in } N_k\}} \|y_r - v_i\|^2$.

The centroid update is

$$v_i = \sum_k u_{ik}^p (y_k + (\alpha / N_R) \sum_r y_r) / ((1 + \alpha) \sum_k u_{ik}^p)$$

Centroids are initialized using k-means on the intensity histogram.

The algorithm alternates between the membership update and the centroid update until convergence. The algorithm halts when the maximum change in any centroid falls below $\epsilon = 0.001$, or after 100 iterations.

2.2 Bias field estimation

The observed intensity is modelled as the product of the true intensity and a smooth multiplicative bias field. Working in the log domain converts this to an additive model. The bias field at voxel k is estimated as

$$\text{beta}_k = y_k - \frac{\sum_i u_{ik}^p v_i}{\sum_i u_{ik}^p}$$

and is smoothed with a mean filter before being subtracted from the log-domain image at each iteration.

3. Preprocessing

Intensities are normalized to the range $[0, 1]$ before clustering. Background voxels are excluded using a simple intensity threshold at the 5th percentile, which prevents the background from dominating the centroid estimates.

4. Failure modes

The method is sensitive to initialization when tissue contrast is low, and the neighbourhood term over-smooths thin structures when α is set too high.